

Part I

1. Request two numbers from the user and print the multiplication of the numbers.

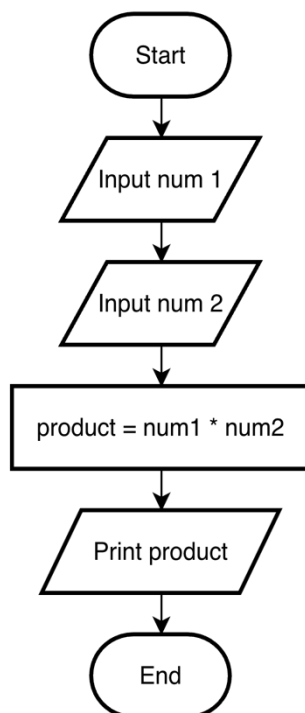
IPO Model

Input	Process	Output
num1, num2	Calculate product by multiplying num1 by num2	product

Pseudocode

1. Start
2. Input num1
3. Input num2
4. $\text{product} = \text{num1} * \text{num2}$
5. Print product
6. End

Flow Chart



2. Determine whether a random number is greater than 50.

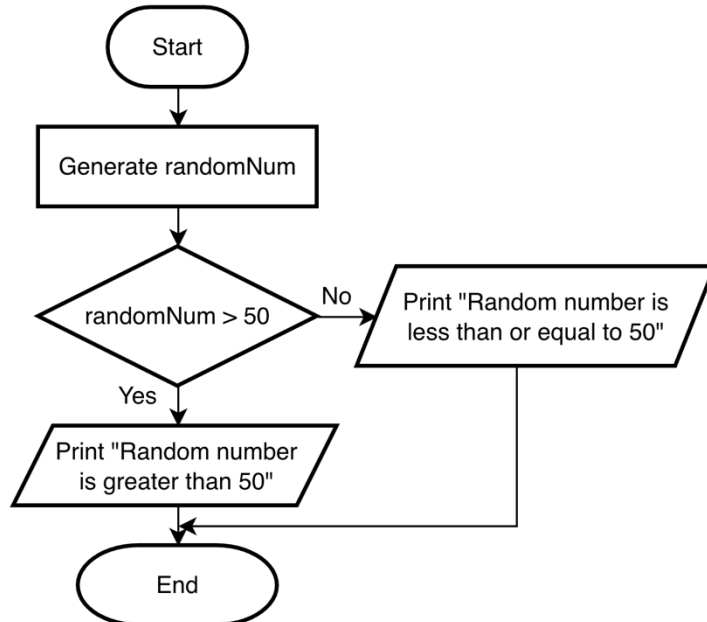
IPO Model

Input	Process	Output
-	1. Generate randomNum 2. Determine whether randomNum is greater than 50	Message "Random number is greater than 50" if random number is greater than 50 or "Random number is less than or equal to 50" if random number is not greater than 50

Pseudocode

1. Start
2. Generate randomNum
3. If randomNum > 50
4. Print "Random number is greater than 50"
5. Else
6. Print "Random number is less than or equal to 50"
7. End

Flow Chart



3. Print the pass/fail grade based on the mark entered by user. The passing mark is at least 40.

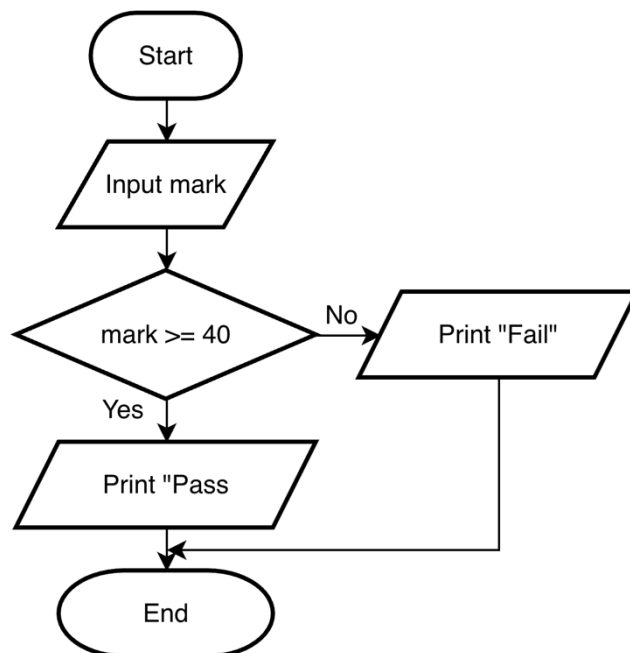
IPO Model

Input	Process	Output
mark	Determine pass/fail grade by checking if mark is greater or equal to 40	Message "Pass" if mark is greater or equal to 40, or "Fail" if mark is less than 40

Pseudocode

1. Start
2. Input mark
3. If mark ≥ 40
4. Print "Pass"
5. Else
6. Print "Fail"
7. End

Flow Chart



4. Print the results of the two players' dice game.

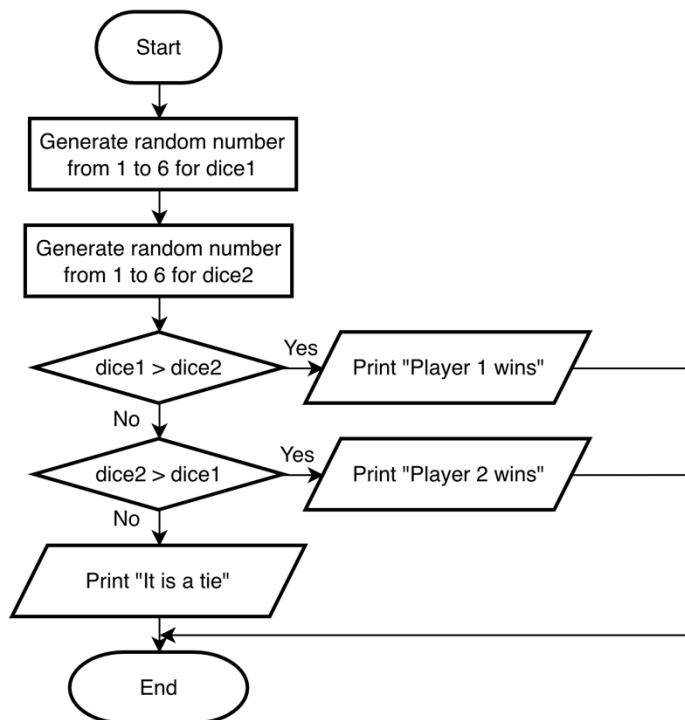
IPO Model

Input	Process	Output
-	1. Generate random number from 1 to 6 for dice1 2. Generate random number from 1 to 6 for dice2 3. Determine result by comparing dice1 and dice2	Message "Player 1 wins" if dice1 is greater than dice2, or "Player 2 wins" if dice2 is greater than dice1, or "It is a tie" if dice1 is equal to dice2

Pseudocode

1. Start
2. Generate random number from 1 to 6 for dice1
3. Generate random number from 1 to 6 for dice2
4. If $\text{dice1} > \text{dice2}$
5. Print "Player 1 wins "
6. Else if $\text{dice2} > \text{dice1}$
7. Print "Player 2 wins"
8. Else
9. Print "It is a tie"
10. End

Flow Chart



5. Print the perimeter of a rectangle.

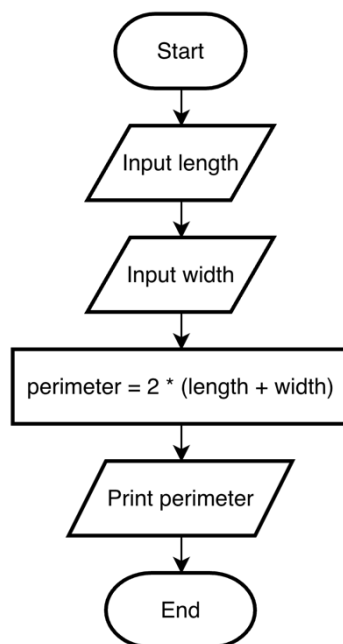
IPO Model

Input	Process	Output
length, width	Calculate perimeter based on length and width	perimeter

Pseudocode

1. Start
2. Input length
3. Input width
4. $\text{perimeter} = 2 * (\text{length} + \text{width})$
5. Print perimeter
6. End

Flow Chart



6. Print the minimum number from 10 random numbers generated by computer.

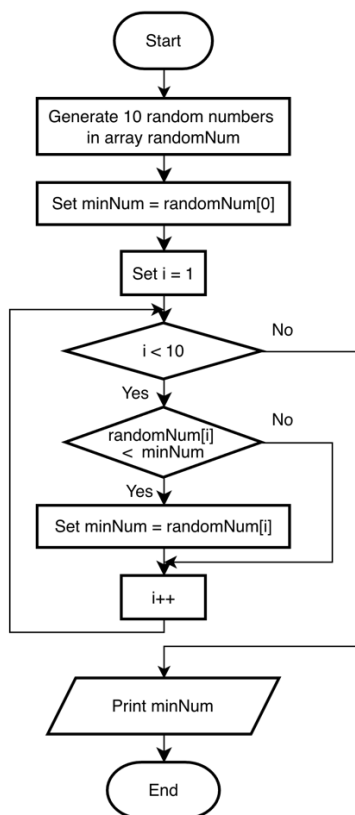
IPO Model

Input	Process	Output
-	1. Generate 10 random numbers in array randomNum 2. Determine minNum among the 10 random numbers	minNum

Pseudocode

1. Start
2. Generate 10 random numbers in array randomNum
3. Set minNum = randomNum[0]
4. Set i = 1
5. For i < 10
6. If randomNum[i] < minNum
7. Set minNum = randomNum[i]
8. i++
9. Print minNum
10. End

Flow Chart



7. Print the number of odd and even number from 10 random numbers generated by computer. The random number must be from 10 – 100.

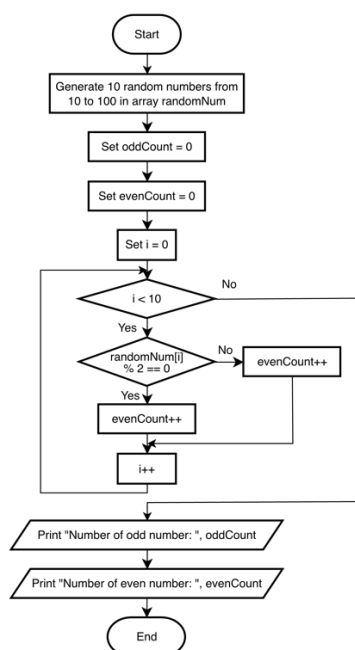
IPO Model

Input	Process	Output
-	1. Generate 10 random numbers from 10 to 100 in array randomNum 2. Count oddCount and evenCount among 10 random numbers	oddCount, evenCount

Pseudocode

1. Start
2. Generate 10 random numbers from 10 to 100 in array randomNum
3. Set oddCount = 0
4. Set evenCount = 0
5. Set i = 0
6. For i < 10
7. If randomNum[i] % 2 == 0
8. evenCount++
9. Else
10. OddCount++
11. i++
12. Print "Number of odd number: ", oddCount
13. Print "Number of even number: ", evenCount
14. End

Flow Chart



Part II

8. Count the number of alphabet U and M from a sentence entered by user.

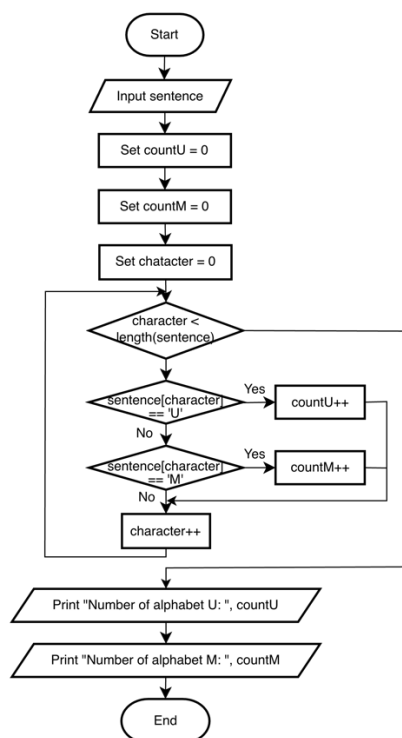
IPO Model

Input	Process	Output
sentence	Count number of alphabet 'U' and 'M' in the sentence	countU, countM

Pseudocode

1. Start
2. Input sentence
3. Set countU = 0
4. Set countM = 0
5. Set character = 0
6. For character < length(sentence)
7. If sentence[character] == 'U'
8. countU++
9. Else if sentence[character] == 'M'
10. countM++
11. character++
12. Print "Number of alphabet U: ", countU
13. Print "Number of alphabet M: ", countM
14. End

Flow Chart



9. Display the frequency of a keyword from a web page.

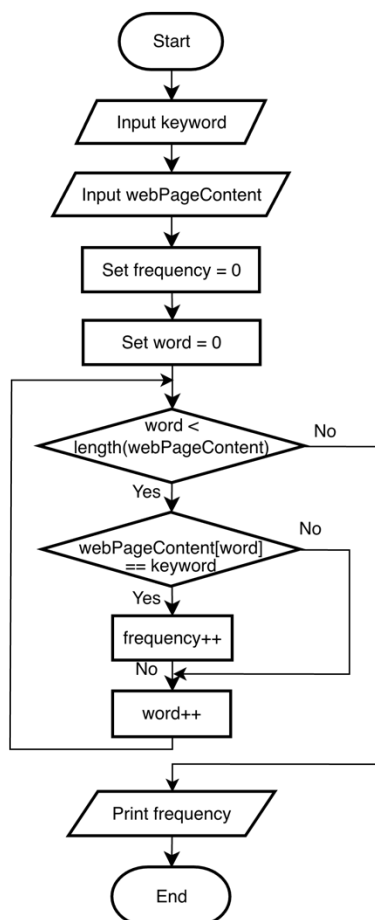
IPO Model

Input	Process	Output
keyword, webPageContent	Count frequency of keyword in webPageContent	frequency

Pseudocode

1. Start
2. Input keyword
3. Input webPageContent
4. Set frequency = 0
5. Set word = 0
6. For word < length(webPageContent)
7. If webPageContent[word] == keyword
8. frequency++
9. word++
10. Print frequency
11. End

Flow Chart



10. Display the number of female student from a random list of 100 students.

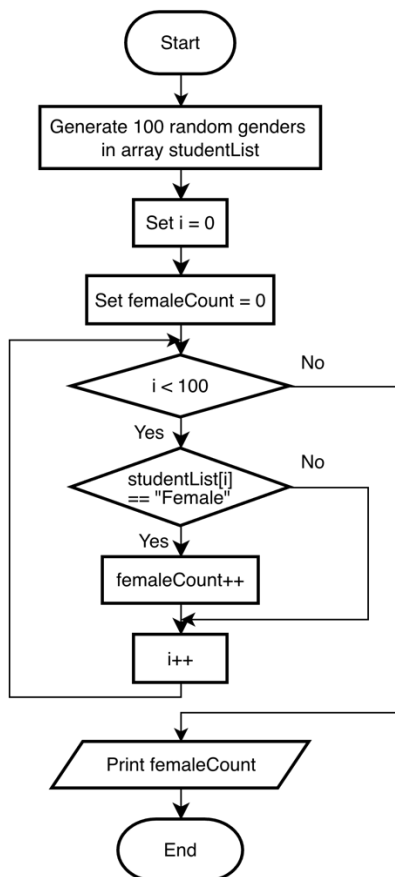
IPO Model

Input	Process	Output
-	1. Generate 100 random students with their gender in array studentList 2. Count femaleCount in studentList	femaleCount

Pseudocode

1. Start
2. Generate 100 random students with their gender in array studentList
3. Set i = 0
4. Set femaleCount = 0
5. For i < 100
6. If studentList[i] == "Female"
7. femaleCount++
8. i++
9. Print femaleCount
10. End

Flow Chart



11. Display a list of 5 random numbers in descending order. (Sort)

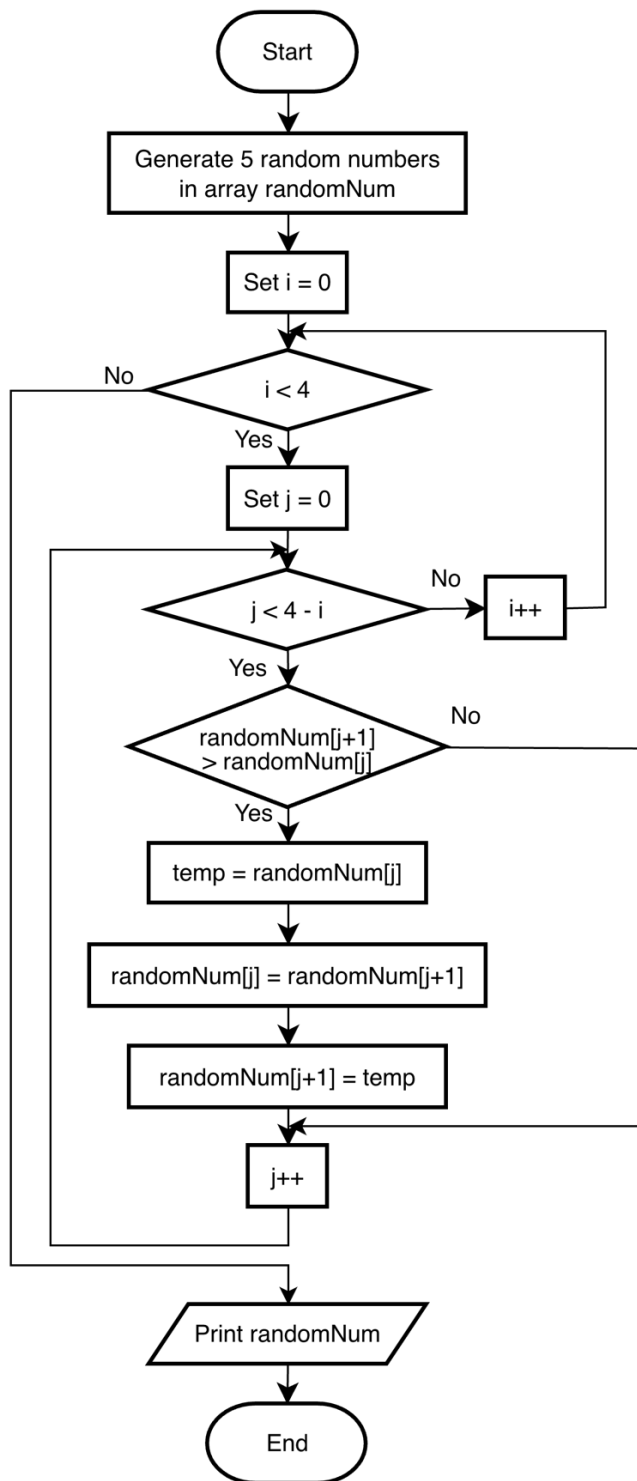
IPO Model

Input	Process	Output
-	1. Generate 5 random numbers in array randomNum 2. Sort 5 random numbers in descending order	List of 5 random numbers in descending order

Pseudocode

1. Start
2. Generate 5 random numbers in array randomNum
3. Set i = 0
4. For i < 4
5. Set j = 0
6. For j < 4 - i
7. If randomNum[j+1] > randomNum[j]
8. temp = randomNum[j]
9. randomNum[j] = randomNum[j+1]
10. randomNum[j+1] = temp
11. j++
12. i++
13. Print randomNum
14. End

Flow Chart



12. Guess a random number generated by computer.

IPO Model

Input	Process	Output
guess	1. Generate randomNum 2. Compare guess with random number to check if it is the same	Message "Correct" if guess is equal to randomNum, or "Too low" if guess is less than randomNum, or "Too high" if guess is greater than randomNum

Pseudocode

1. Start
2. Generate randomNum
3. Input guess
4. If guess == randomNum
5. Print "Correct"
6. Else if guess < randomNum
7. Print "Too low"
8. Else
9. Print "Too high"
10. End

Flow Chart

